

## OUESSANT-STIFF, France

WMO: 071000

Lat: 48.4733N

Lon: 5.0569W

Elev: 223

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 36.4       | 38.7 | 27.2                          | 21.3 | 39.2 | 29.7 | 24.0 | 40.7 | 49.9                  | 47.4 | 45.5 | 47.5 | 23.0                  | 100  | 0.800 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.5                    | 69.6            | 63.6 | 67.6 | 62.8 | 66.2 | 61.9 | 65.0                | 67.6 | 63.8 | 66.2 | 62.9 | 65.0 | 12.1                 | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 63.9                            | 89.7 | 65.8 | 62.9 | 86.4 | 64.9 | 61.9 | 83.4 | 64.0 | 30.1          | 67.8 | 29.2 | 66.4 | 28.5 | 65.1 | 71.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 41.3          | 36.8 | 33.2 | DB                         | 32.8 | 74.8               | 3.9 | 3.1                                                | 30.0 | 77.0       | 27.8 | 78.8       | 25.6 | 80.5       | 22.8 | 82.8 | (4) |
| (5)               |      |      | WB                         | 30.1 | 67.6               | 3.5 | 1.7                                                | 27.6 | 68.8       | 25.5 | 69.8       | 23.5 | 70.7       | 20.9 | 71.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 54.1        | 47.8  | 47.1  | 48.7  | 50.8  | 54.4  | 58.3  | 61.4  | 61.9  | 59.9  | 56.7  | 52.3  | 48.9  |  |  |
|                                                                        | DBStd       | 6.11        | 4.04  | 3.76  | 3.30  | 2.98  | 2.75  | 2.61  | 2.36  | 1.97  | 2.46  | 2.97  | 3.32  | 4.10  |  |  |
|                                                                        | HDD50       | 346         | 86    | 89    | 62    | 24    | 2     | 0     | 0     | 0     | 0     | 1     | 15    | 67    |  |  |
|                                                                        | HDD65       | 4000        | 534   | 500   | 507   | 427   | 328   | 201   | 115   | 98    | 154   | 257   | 381   | 498   |  |  |
|                                                                        | CDD50       | 1829        | 17    | 9     | 20    | 47    | 139   | 250   | 353   | 369   | 297   | 209   | 85    | 34    |  |  |
|                                                                        | CDD65       | 7           | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 2     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 11          | 0     | 0     | 0     | 0     | 0     | 2     | 5     | 3     | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 17.4        | 21.8  | 20.7  | 19.2  | 17.4  | 15.7  | 14.0  | 13.5  | 13.1  | 14.1  | 16.9  | 20.7  | 21.9  |  |  |
|                                                                        | PrecAvg     | 29.8        | 3.7   | 3.0   | 2.0   | 2.2   | 1.9   | 1.5   | 1.9   | 1.9   | 1.9   | 3.0   | 3.5   | 3.3   |  |  |
| (15) Precipitation                                                     | PrecMax     | 40.1        | 7.5   | 7.4   | 4.9   | 5.4   | 4.4   | 3.7   | 4.4   | 5.1   | 5.0   | 4.3   | 7.4   | 6.0   |  |  |
|                                                                        | PrecMin     | 20.9        | 0.7   | 0.4   | 0.4   | 0.3   | 0.2   | 0.4   | 0.2   | 0.1   | 0.2   | 0.6   | 0.6   | 0.4   |  |  |
|                                                                        | PrecStd     | 4.6         | 1.8   | 1.6   | 1.1   | 1.4   | 1.2   | 0.9   | 1.1   | 1.3   | 1.2   | 1.0   | 1.8   | 1.5   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 55.7  | 55.4  | 58.4  | 62.5  | 66.0  | 70.9  | 72.5  | 72.0  | 70.7  | 65.5  | 60.8  | 57.5  |  |  |
|                                                                        |             | MCWB        | 54.0  | 52.6  | 53.2  | 55.7  | 60.1  | 63.5  | 65.0  | 64.8  | 63.9  | 61.3  | 58.5  | 55.7  |  |  |
|                                                                        | 2%          | DB          | 54.7  | 53.9  | 55.3  | 58.3  | 61.9  | 66.8  | 69.4  | 68.9  | 67.3  | 63.3  | 59.2  | 56.5  |  |  |
|                                                                        |             | MCWB        | 53.3  | 52.0  | 52.0  | 53.9  | 57.7  | 61.4  | 63.5  | 63.9  | 62.8  | 60.3  | 57.1  | 55.1  |  |  |
|                                                                        | 5%          | DB          | 53.8  | 52.9  | 53.9  | 56.5  | 60.1  | 64.5  | 67.4  | 67.3  | 65.4  | 61.9  | 58.1  | 55.6  |  |  |
|                                                                        |             | MCWB        | 52.5  | 51.3  | 51.3  | 52.8  | 56.4  | 60.2  | 62.5  | 62.8  | 61.5  | 59.5  | 56.1  | 54.3  |  |  |
|                                                                        | 10%         | DB          | 52.8  | 52.0  | 52.8  | 55.0  | 58.6  | 62.8  | 65.8  | 65.9  | 63.9  | 60.7  | 57.0  | 54.6  |  |  |
|                                                                        |             | MCWB        | 51.4  | 50.5  | 50.6  | 51.9  | 55.3  | 59.1  | 61.7  | 62.0  | 60.6  | 58.7  | 55.2  | 53.2  |  |  |
|                                                                        | 0.4%        | WB          | 54.9  | 53.8  | 54.4  | 56.9  | 60.9  | 64.5  | 66.3  | 66.9  | 65.7  | 62.7  | 59.6  | 56.7  |  |  |
|                                                                        |             | MCDB        | 55.1  | 54.3  | 56.2  | 60.4  | 64.4  | 68.7  | 70.6  | 69.6  | 68.3  | 64.2  | 59.9  | 56.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 53.9  | 52.8  | 53.1  | 54.8  | 58.4  | 62.3  | 64.6  | 65.1  | 63.9  | 61.2  | 57.8  | 55.7  |  |  |
|                                                                        |             | MCDB        | 54.4  | 53.3  | 54.4  | 57.4  | 61.0  | 65.5  | 67.7  | 67.2  | 66.0  | 62.4  | 58.5  | 56.1  |  |  |
|                                                                        | 5%          | WB          | 52.9  | 51.9  | 52.1  | 53.6  | 57.0  | 60.9  | 63.4  | 64.0  | 62.6  | 60.2  | 56.8  | 54.7  |  |  |
|                                                                        |             | MCDB        | 53.5  | 52.5  | 53.3  | 55.6  | 59.3  | 63.6  | 66.1  | 65.9  | 64.3  | 61.3  | 57.7  | 55.3  |  |  |
|                                                                        | 10%         | WB          | 51.9  | 50.9  | 51.2  | 52.5  | 56.0  | 59.7  | 62.4  | 63.0  | 61.4  | 59.3  | 55.5  | 53.7  |  |  |
|                                                                        |             | MCDB        | 52.6  | 51.8  | 52.4  | 54.4  | 58.0  | 62.0  | 64.8  | 64.8  | 63.1  | 60.3  | 56.8  | 54.5  |  |  |
|                                                                        | 5% DB       | MDBR        | 5.3   | 5.9   | 6.5   | 7.5   | 7.8   | 8.1   | 8.5   | 8.5   | 7.9   | 6.5   | 5.3   | 5.3   |  |  |
|                                                                        |             | MCDBR       | 5.5   | 6.3   | 7.6   | 9.4   | 9.5   | 10.6  | 10.7  | 10.2  | 10.0  | 7.5   | 5.9   | 5.1   |  |  |
|                                                                        |             | MCWBR       | 6.0   | 6.0   | 6.1   | 6.6   | 6.6   | 7.1   | 6.6   | 6.6   | 6.5   | 5.7   | 5.7   | 5.9   |  |  |
|                                                                        |             | MCWBR       | 5.4   | 5.8   | 6.7   | 8.4   | 8.8   | 9.7   | 9.7   | 8.9   | 9.0   | 6.7   | 5.4   | 5.0   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR       | 6.4   | 6.3   | 6.1   | 6.4   | 6.5   | 6.9   | 6.7   | 6.7   | 6.6   | 5.7   | 6.0   | 6.1   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        | taub        |             | 0.340 | 0.353 | 0.379 | 0.384 | 0.387 | 0.379 | 0.382 | 0.371 | 0.363 | 0.360 | 0.340 | 0.332 |  |  |
|                                                                        |             | taud        | 2.391 | 2.385 | 2.337 | 2.348 | 2.384 | 2.435 | 2.432 | 2.484 | 2.487 | 2.464 | 2.424 | 2.358 |  |  |
| (42) Clear-Sky Solar Irradiance                                        | Ebn at Noon |             | 226   | 252   | 264   | 274   | 278   | 280   | 277   | 276   | 267   | 247   | 225   | 211   |  |  |
|                                                                        |             | Edh at Noon | 22    | 27    | 33    | 36    | 36    | 35    | 35    | 31    | 29    | 25    | 21    | 20    |  |  |
|                                                                        | RadAvg      |             | 304   | 559   | 937   | 1473  | 1747  | 1887  | 1792  | 1536  | 1224  | 721   | 382   | 253   |  |  |
| (45) All-Sky Solar Radiation                                           | RadStd      |             | 22    | 58    | 96    | 121   | 154   | 197   | 97    | 126   | 87    | 52    | 30    | 30    |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (21 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page